

Statistical & ML Model Deployment — Beginner Cheat Sheet

Core lifecycle

DATA → TRAIN → VALIDATE → SERIALISE → DEPLOY → LOAD → INFERENCE → MONITOR → RETRAIN

1. Training vs Deployment

Stage	What it means
Training	Learn model parameters from historical data.
Validation	Check whether the model works well on unseen data.
Serialization	Save the fitted model so it can be loaded later.
Deployment	Make the model available to production systems.
Inference	Use the trained model on new data.
Monitoring	Watch errors, data changes, and model performance.
Retraining	Create a new model version when the old one becomes unsuitable.

2. Multiple Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Intercept (β_0) — predicted value when all inputs are zero.

Coefficient / slope (β) — how much the prediction changes when that variable increases by 1, holding other variables constant.

Residual — actual value – predicted value. It is the model's prediction error for an observation.

3. What to Serialize

For prediction	For statistical inference
Intercept	Intercept
β coefficients	β coefficients
Feature names and order	Standard errors
Preprocessing parameters	p-values
Model/version metadata	Confidence intervals
	t-statistics
	Covariance matrix
	Degrees of freedom
	Residual statistics / diagnostics

4. Statistical Terms — Plain English

Term	Beginner meaning
Standard error	How uncertain / imprecise an estimated coefficient is.
p-value	How surprising the observed evidence would be if there were really no effect.
Confidence interval	A range expressing uncertainty around an estimated coefficient.
t-statistic	Coefficient compared with its standard error; roughly $\beta \div \text{SE}$.

Residual statistics	Summaries of prediction errors, such as RMSE and residual patterns.
Covariance matrix	Shows coefficient uncertainty and how coefficient estimates move together.
Degrees of freedom	Roughly, how much independent information remains for estimating uncertainty after fitting parameters.

5. Can You Save Only β ?

Yes, if the goal is only prediction and you also preserve the intercept, exact feature order/definitions, and every preprocessing step needed to transform new inputs exactly as during training.

Example artifact: { model_type, version, features, intercept, coefficients, preprocessing }. JSON is often a good choice for a simple linear model because it is small, inspectable, and language-independent.

6. Clustering Deployment

For K-Means, training learns cluster centroids. Serialize the fitted model (or centroids plus preprocessing). In production, preprocess new data the same way, then assign each observation to its nearest learned cluster.

7. API vs Batch Deployment

Online/API: Client → REST API → preprocessing → model → prediction → response. Use when an immediate prediction is needed.

Batch: Database → scheduled job → load model → predict many records → save results. Use for large scheduled workloads.

8. Golden Rule

Training learns parameters. Serialization saves the fitted model. Deployment makes it available. Inference uses it. Monitoring tells you when it needs attention.

Most common production mistake: training and production preprocessing do not match. Keep the preprocessing definition and parameters with the model artifact.